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PMC6222659 | Suzuki – Miyaura 偶联 | 采集状态 imported_unchecked

标题	Efficient Suzuki – Miyaura C-C Cross-Couplings Induced by Novel Heterodinuclear Pd-bpydc-Ln Scaffolds
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出处	Molecules : A Journal of Synthetic Chemistry and Natural Product Chemistry 2018
DOI	10.3390/molecules23102435
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC6222659/
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摘要（原文节选）

An easy access to a series of previously unreported heterodinuclear Pd-Ln compounds, Pd-bpydc-La, Pd-bpydc-Ce and Pd-bpydc-Nd (bpydc = 2,2 ′ -bipyridine-5,5 ′ -dicarboxylate) has been developed. The Pd-Ln hybrid networks were effectively applied as catalysts in Suzuki – Miyaura C-C cross-coupling reactions of 4-bromoanisole and 4-bromobenzonitrile with phenylboronic acid, under mild conditions. A systematic investigation revealed Pd-bpydc-Nd as the most active catalyst. In all cases, reaction yields varied with the base, ⋯

数据表 · Table 1（12 行，含数值行 12 行）

Suzuki – Miyaura cross-coupling of 4-bromoanisole with phenylboronic acid in methanol-induced by Pd-bpydc-Ln catalysts 1 – 3.

Entry	Catalyst	T (° C)	Time (h)	Yield (%) a,b,c
1	Pd-bpydc-La (1)	30	8	35.0 a
2	30	8	15.0 b	
3	60	4	55.0 a	
4	60	4	14.9 b	
5	Pd-bpydc-Ce (2)	30	8	25.0 a
6	30	8	25.9 b	
7	60	4	91.9 a	
8	60	4	88.9 b	
9	Pd-bpydc-Nd (3)	30	8	95.0 a
10	30	8	93.4 b	
11	60	4	95.0 a	
12	60	4	94.9 b	

脚注（逐字）：a Reaction conditions: 4-bromoanisole (0.5 mmol), phenylboronic acid (0.6 mmol), sodium tert-butoxide (0.5 mmol), methanol (1 mL),catalyst (0.5 mol %); b Catalyst (0.2 mol %).; c Reaction yield (%) for total conversion determined from the internal standard yield, at the specified reaction time, base

数据表 · Table 3（5 行，含数值行 5 行）

Suzuki – Miyaura cross-coupling of 4-bromoanisole with phenylboronic acid in the presence of Pd-bpydc-Nd (3) when various bases are employed a.

Entry	Catalyst	Base	Yield (%) b
1	Pd-bpydc-Nd(3)	KOH	95.0
2	K2CO3	91.9	
3	NaOH	92.0	
4	Na2CO3	92.0	
5	NaOtBu	95.0	

脚注（逐字）：a Reaction conditions: 4-bromoanisole (0.5 mmol), phenylboronic acid (0.6 mmol), base (0.5 mmol), methanol (1 mL); reaction time 8 h; temperature: 30 ° C, catalyst: 0.2 mol %. b Reaction yield (%) for total conversion determined from the internal standard yield, at the specified reaction time, based

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